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Investigation of the development of the Robotic Dexterous Hands

Challenges & Difficulties, potential Develop and Analysis aspects

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Summary of last PowerPoints:

The synergy of Mechanics, Transmission, and Actuation forms the indispensable physical basis for dexterous hand intelligence

Mechanical

- Adaptability
- Coupling
- Rigidity

Transmission

- Tendo-Drive
- Linkage-Driven

Actuation

- Rotary Motors
- Linear Servos
- Worm Gear

Market validation

Driven by the booming humanoid robot market (CAGR > 26%), optimizing this hardware stack has never been more critical

Future Challenges

The key challenge lies in balancing performance, cost, and reliability, which highlights the need for continuous innovation in modularization and robust design



Comparison between tendon-driven and linkage/gear-driven

Aspects	Core Principle	Actuator Placement	Precision & Stiffness	Back drivability	Force/Tactile Sensing	Inherent Compliance & Safety	Complexity & Maintenance	Typical Applications
Tendon-Driven	Transmits force and motion remotely via cables (tendons), similar to biological tendons	Actuators can be remotely located, drastically reduction the weight of the end-effector	Lower, easy to elastic stretch, friction, hysteresis, and lower overall stiffness	Generally poor	Joint torque can be estimated indirectly by measuring tendon tension, through friction can corrupt the signal	Inherently compliant, elasticity of tendons can absorb shocks, making interactions safer	Complex, require frequent maintenance	Dexterous hands, lightweight robot arms, bio-inspired robots
Linkage/Gear-Driven	Transmits force and motion directly through rigid components (gears, linkages)	Actuators are typically mounted at or near the joint, increasing the weight and inertia of the moving parts	High, offers high rigidity, high transmission efficiency, low backlash, and excellent positional accuracy	Generally good	Typically requires force/torque sensors to be mounted directly on the joint, adding complexity	Inherently rigid, achieving compliance requires complex active force control algorithms	Structurally complex but highly reliable, requires less frequent maintenance	Industrial robot arms, high-precision CNC machines, articulated robots



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Technical bottlenecks and industrialization difficulties:

1. The efficiency and reliability of the transmission system
2. High-density sensing and integration
3. Cost-performance trade-off

Potential research directions and development areas:

- I. Modular and standardized design
- II. Soft-robot fusion
- III. New transmission and drive solutions
- IV. Bionic design and smart materials



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Modular and standardized design Case Study

高可靠性

>150,000

次的使用寿命



低功耗

姿态保持
不耗电



高自由度

19 DOF

12个主动自由度7个被动自由度



强兼容

支持主流机器人
末端接口和数据格式



多模态

包含可感知4个模态
位置、力觉、滑觉、接近觉



易维护

5根手指均可

独立更换

高集成

集成多种通信接口
支持灵巧手触觉算法的

二次开发

DexRobot
灵巧智能

强大的指尖力量

单指

1,000g

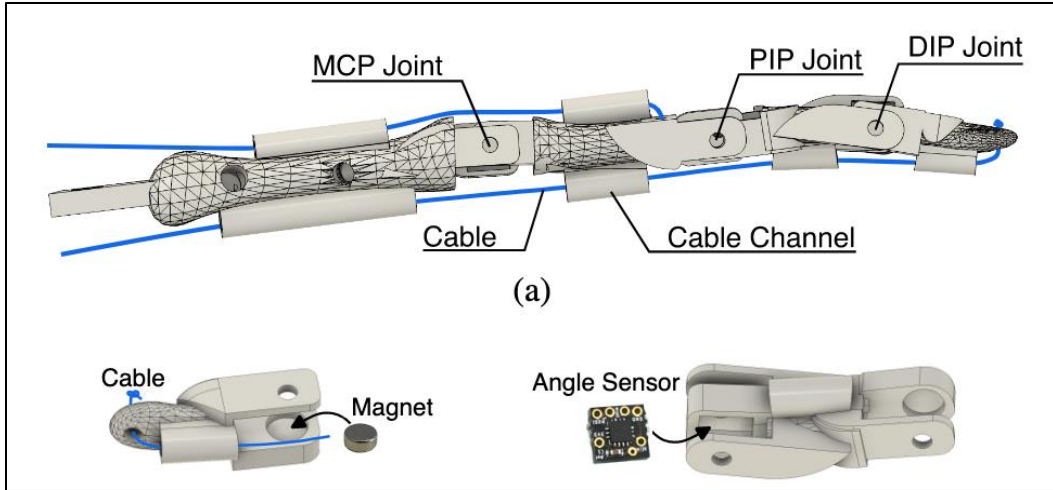
负载力



Drive method	Linear Motor + Tendon Drive
weight	1.0 kg
Fan cooling capacity	2.38cfm
sensors	Location, tactile, force



Soft-robot fusion Case Study



combining the **inherent flexibility** and **safety** of soft robots with the **precision** and **strength** of rigid structures to develop a flexible and rigid-coupled dexterous hand may be an effective way to break through the current difficulty in balancing adaptability and rigidity.

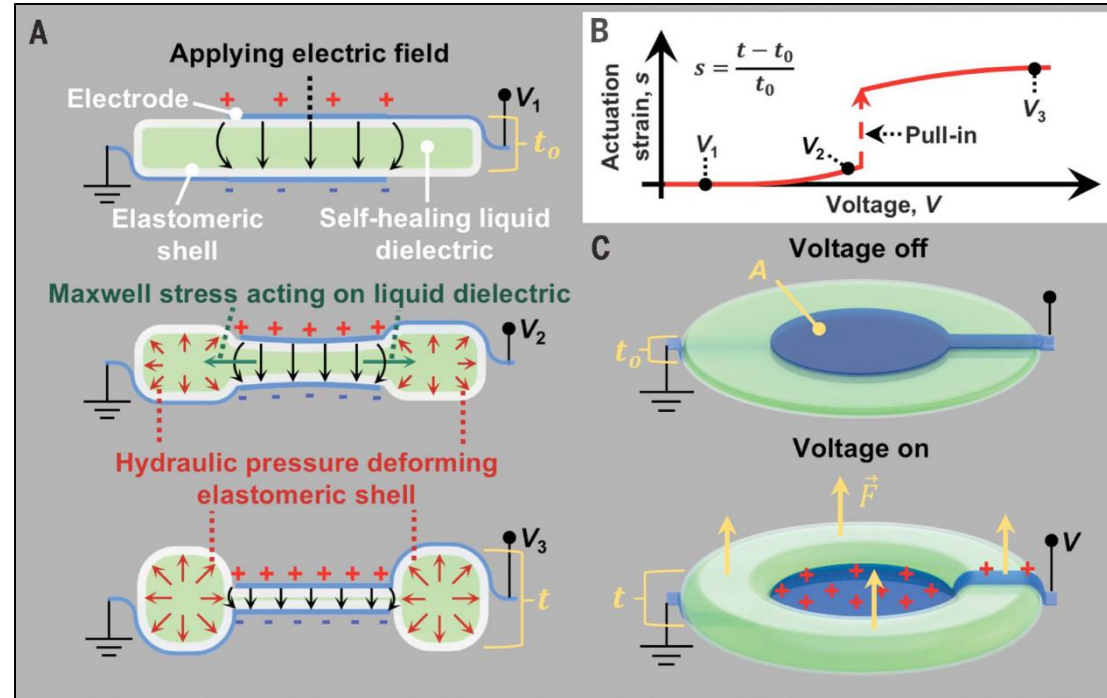
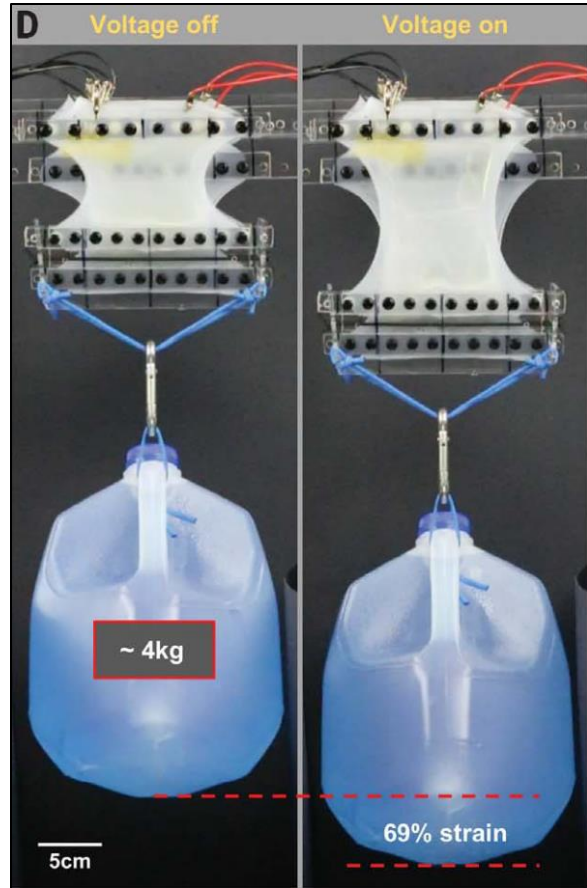




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New transmission and drive solutions & Bionic design and smart materials

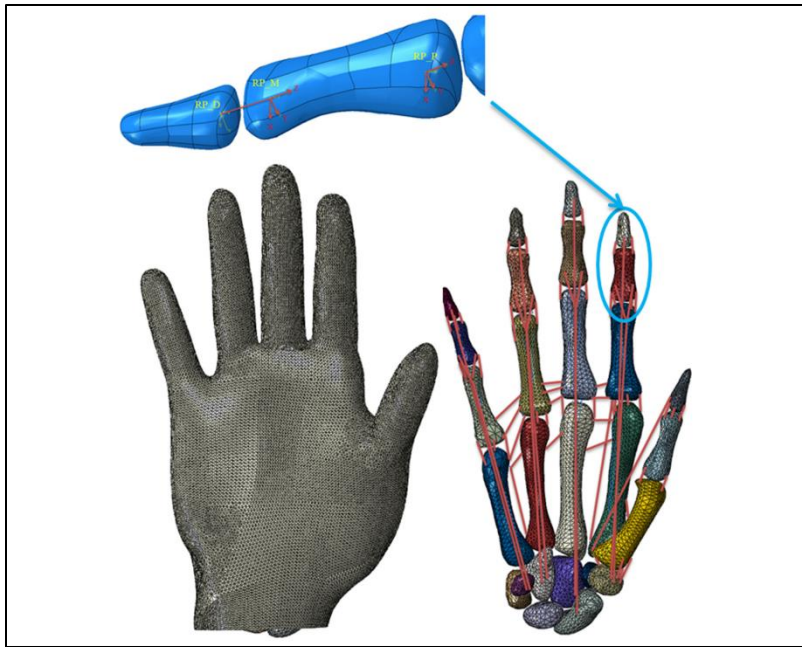


Low-friction, high-rigidity Composite transmission mechanisms to fundamentally improve transmission efficiency, weight and reliability



New transmission and drive solutions & Bionic design and smart materials

Case Study



Kinematic motion definition and anatomical view of the FE hand model. The red lines represent ligaments and tendons. The diagram on the top shows the boundary conditions defined at each interphalangeal joint.



Finger Extensor Mechanism

- Beyond simple tendon drives
- Embrace Physical Intelligence
- A passive, web-like structure



- Stiffness is not the goal
- Use moderately stiff & durable materials
- Performance saturates; avoid over-engineering



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Summery

Potential research directions & development areas

Modular and standardized design

Soft-robot fusion

What are the Requirements?

- Working environment
- Movements involved
- Weight limit
- Number of DOF
- Etc...



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